

HIPHOP-MOVE RECOGNITION WITH A MARKERLESS MoCAP SYSTEM AND A SIMPLE 1D-CNN

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ABSTRACT

This paper presents a tool that uses camera-based, markerless Motion Capture (MoCap) system and a simple 1D Convolutional Neural Network (1D-CNN) for automatic recognition of a small vocabulary consisting classic hiphop moves. The MoCap system used is MediaPipe's Pose Landmarker task. This approach showed great results during test achieving an accuracy of 0.95, and additionally overall good performance in a real-time demo. Potential applications include automatic annotation of hiphop segments in longer videos and educational feedback systems.

1. INTRODUCTION

Hiphop dance is more than a fixed list of movements. It consists of a wide range of styles and individual expressions. Still, there exists a set of classic hiphop moves that are widely recognized and characteristic, such as the Dougie and the running man [1]. Automatically annotating such movements can be useful for building datasets for training models, evaluating dancing robots, providing feedback to learners, and annotating educational video material that includes hiphop dance. Manual annotation of movement is time-consuming and tedious, which motivates the development of automatic tools.

To the best of my knowledge, there are currently no tools that automatically annotate hiphop dance movements from motion capture or video data. This project explores a possible approach. The MediaPipe Pose Landmarker task is used as a camera-based, markerless MoCap system, and tracked 3D joint trajectories form the core representation of hiphop movements. These MoCap features are then combined with a small-scale 1D-CNN model to investigate questions such as: "Can a small vocabulary of six hiphop dance moves be automatically recognized from camera-based, markerless MoCap-data using simple, low-dimensional and human-interpretable features and a simple model, and what artistic or pedagogical uses could such a movement classifier have?"

2. RELATED WORKS

MoCap systems are widely used in robotics, sports and entertainment [2]. In this project, a MoCap system is combined with machine learning to classify dance movements,

building on existing work that uses different systems and technologies.

The system presented by Simpson et al. could recognize different ballet- and Tai Chi Chuan moves based on lower body movement [3]. At its best, the accuracy of the system was 97%. Their dataset included a professional ballet dancer and a Tai Chi Chuan expert. A motivation for their work was the lack of existing automatic annotation of movements for movies.

Gong and Du used IMU-based MoCap and a simplified 10-node skeleton to classify dance movements, achieving F1-scores over 0.7 [4].

Lu et al. showed how a Spatial Temporal Graph Convolutional Network (ST-GCNs) with skeletal information outperformed other ST-GCNs in recognizing dance movements with Kinect v2 [5].

Another article that used the Kinect (not v2), developed an Extreme Learning Machine classifier which is a type of Neural Network (NN) with one single layer [6]. They used Rectified Linear Unit (ReLU) activation and reduced the high-dimensional data with Principal Component Analysis and Linear Discriminant Analysis [6].

These systems rely on optical- or depth-based MoCap to obtain skeletal joint trajectories for dance-movement recognition [3] [4] [5] [6]. While effective, such setups generally require specialized hardware and controlled recording environments, which limits accessibility compared to an ordinary camera. In contrast, this work uses a camera-based, markerless MoCap approach via MediaPipe [7], prioritizing accessibility, ecological validity and low cost. This choice allows us to investigate how far simple 3D joint trajectories from a single camera can go towards reliable hiphop-move recognition.

3. SYSTEM DESIGN

The system developed for this project automatically recognizes six hiphop dance-moves with the help of a markerless MoCap system. Figure 1 shows the pipeline: videos are recorded, processed (trimming, segmentation, feature extraction and pre-processing), then used for model training, testing, and a real-time demo.

The MediaPipe Pose Landmarker task was the markerless MoCap system used for this project. The main reasons for choosing this system is accessibility, ecological validity and cost. With only a camera, videos can be collected in many environments without strict setup rules. Markerless systems provides a filming scenario which is closer to the natural state of the movement. Markerless MoCap is less invasive [8], and less equipment disturb the subjects' movements when measured.

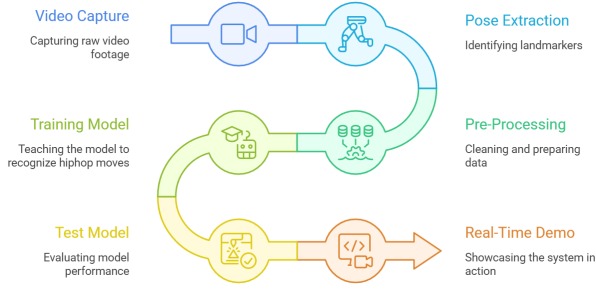


Figure 1. The project pipeline

Label	Hiphop Move
0	Bart Simpson
1	Cabbage Patch
2	Kick-Ball Change
3	Pas de Bourré
4	Steve Martin
5	Wu-Tang

Table 1. The chosen movements and their corresponding label/class.

The task outputs 33 landmarks/skeletal key-points in the 3D (x, y, z) [7]. Only 16 of the 33 landmarks were chosen to be processed further. It was a decision based on visual inspection and dance experience. Inspecting videos with joints and skeleton drawn on top, the face landmarks and finger/hand landmarks seemed redundant to include, because it seemed like the remaining 16 landmarks already provided sufficient information for classifying the moves. Based on personal experience, the head and the fingers/hand does significantly contribute to the standard execution of the dance-moves.

To achieve translation- and scale-invariance, centralization and normalization is necessary [9]. Previous research has scaled the data based on the length of the spine [10]. In this work the shoulder width is used to scale data in each frame.

The model developed for this project was a simple 1D-CNN. A 1D-CNN is an NN that performs convolution operations on one-dimensional data such as timeseries [11]. While 2D or 3D CNNs could be used, the goal here is to reduce complexity and explore smaller architectures for this classification problem. Therefore the videos are processed to one-dimensional timeseries data of the extracted landmarks. An advantage of 1D-CNNs is that they require fewer operations than 2D-CNNs which makes them faster and more suitable for real-time applications on mobile devices [12].

The activation function ReLU is used because of its property of avoiding the vanishing gradient problem [13]. Global Average Pooling (GAP) is an operation that takes the average of all the values in each feature map and puts it into one single vector [14]. Using GAP reduces the amount of parameters and mitigates overfitting [15] [16].

A lightweight real-time demo was implemented to evaluate the model during live webcam input. Recent frames are kept in a fixed-size buffer, and once enough frames are collected, the system generates continuous predictions of the current dance-move.

4. METHODS

The six moves that were chosen to be part of the hiphop move vocabulary were Bart Simpson (BS), Cabbage Patch (CP), Kick-Ball Change (KBC), Pas de Bourré (PdB), Steve Martin (SM) and Wu-Tang (WT) (see Table 1). The objectives for the project were as following:

- Data collection
- Data processing
- Model development
- Evaluation

4.1 Data collection

For the data collection, videos were recorded with a smart-phone (Samsung Galaxy S25) and webcam (Lenovo Yoga Slim 7 Pro) 62 videos were recorded in total: 9 of BS and CP, 10 of KBP and SM, 11 of PdB, and 13 of WT. The videos were recorded with a frame-rate of 30 frames per second. The subject was mainly the author, which has 10 years of experience with dancing hiphop. As an attempt to get the model to generalize more, an additional dancer was added to the dataset. The dancer is a non-expert, and seven videos were recorded of this dancer: one per move, except WT which has two videos.

Recordings were made in a simple, frontal full-body setup at 30 fps. For each clip, the dancer repeatedly performed one move at approximately four-count tempo. The videos (15-60s) were thereafter manually labeled and went through four stages of processing.

4.2 Data processing

The first processing stage was trimming the videos. To increase the data quality, the unnecessary parts were removed. These are non-dancing parts that captured walking towards or from the camera. The library Musical Gestures Toolbox for Python was used for trimming the videos [17].

In the second processing stage, the trimmed videos were segmented into clips of approximately four seconds using OpenCV. In total, 369 clips were produced. In total, 369 clips were produced.

In the third stage of processing, the 3D joint trajectory-data were extracted from all of the clips and written into one large csv-file. The landmarks that were extracted in the MediaPipe indexation were number 11 to 16 and 23 to 32. Which joints they represent can be seen in Table 2. Figure 2 shows a plot of the landmarks from one videoframe and illustrates the landmarks/joints that were used.

Among the available MediaPipe Pose Landmarker variants, the Lite model was chosen because it provided sufficiently accurate landmarks while using less memory, which is beneficial for real-time use.

Index	Joint
11	Left Shoulder
12	Right Shoulder
13	Left Elbow
14	Right Elbow
15	Left Wrist
16	Right Wrist
23	Left Hip
24	Right Hip
25	Left Knee
26	Right Knee
27	Left Ankle
28	Right Ankle
29	Left Heel
30	Right Heel
31	Left Foot Index
32	Right Foot Index

Table 2. The MediaPipe indexation for the landmarks and what they represent.

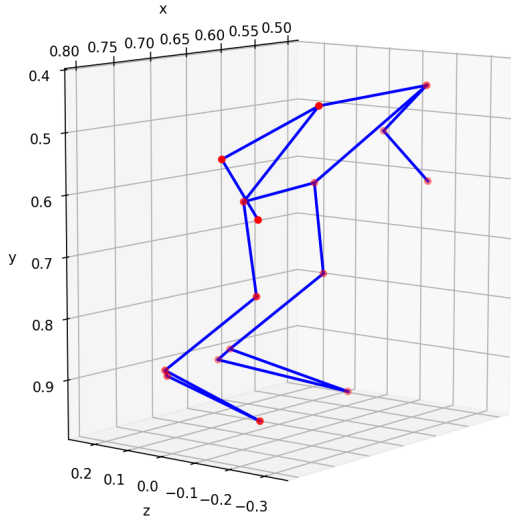


Figure 2. Plot of the landmarks used for this project

In the last processing stage, the data were split on the video level into training, validation and test sets to avoid data leakage between them. For each set, clips were converted into timeseries using overlapping sequence windows of 90 frames and 48 features (16 landmarks in 3D). The window length was chosen to capture at least one iteration of the move despite tempo variations, and windows overlapped by 60 frames.

There were some moves that were subject to occlusion, which lead to missing landmarks. To tackle this, linear interpolation was implemented. For centering the data, origo was defined at the mid-hip and is defined by Equation (1).

$$O = \frac{\text{left hip} + \text{right hip}}{2} \quad (1)$$

After that, the 3D MoCap-data was normalized after the shoulder-width. Now the data has been processed into

timeseries-data which can be used for training and testing the model.

4.3 Model development

The model takes timeseries as input. Figure 3 shows the structure of the model. The input-data has the shape (90, 48) which is fed to a 1D-CNN layer with 16 filters of size 5. This produces feature maps which goes through a GAP layer, and lastly the output layer outputs the vector of six probabilities for the hiphop moves. ReLU is the activation function used in the 1D-CNN layer. The loss used is Categorical Cross-Entropy loss. The optimizer used is Stochastic Gradient Descent. Early stopping is implemented to prevent overfitting. The train-, val-, test-split is 60%, 20% and 20%, respectively. The maximum amount of epochs run is 100 and the batch size is 8.

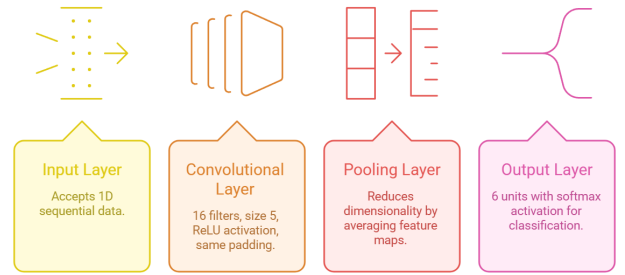


Figure 3. The structure of the 1D-CNN

4.4 Evaluation

The testset was fed into the model to evaluate its performance on unseen data. The confusion matrix was produced after testing. However, the main evaluation happened with the real-time demo.

5. RESULTS

Figure 5 and 4 shows the loss and accuracy after training the model. The model was able to recognize the six hiphop moves with an accuracy of 0.95. The values in Table 3 are calculated with the values in Table 4. Precision, recall and F1-score all have an upper bound at 1.00. The precision is high overall with 0.87 (KBC) as the lower bound. The recall has the lower bound at 0.69 for PdB. Looking at Table 4, we can see that the model sometimes confuses the move PdB with KBC. This also happened when testing in the real-time demo. Although there was some confusion between these two moves at times, the real-time performance was good overall.

6. DISCUSSION

This project investigated whether a small vocabulary of six hiphop moves could be automatically recognized from camera-based, markerless MoCap-data. The system represents each movement as 3D trajectories of 16 selected joints (shoulders, elbows, wrists, hips, knees, ankles, heels

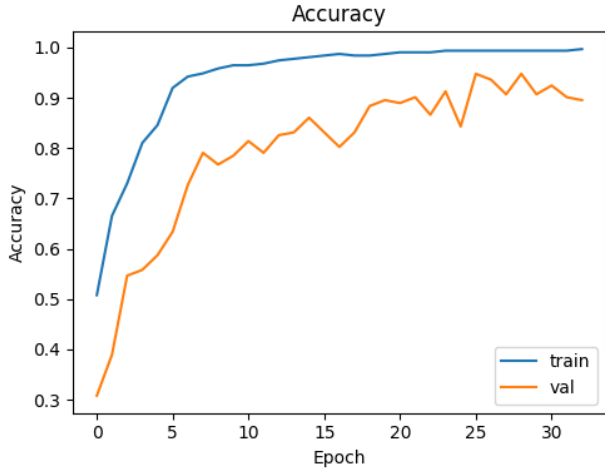


Figure 4. Training and validation accuracy

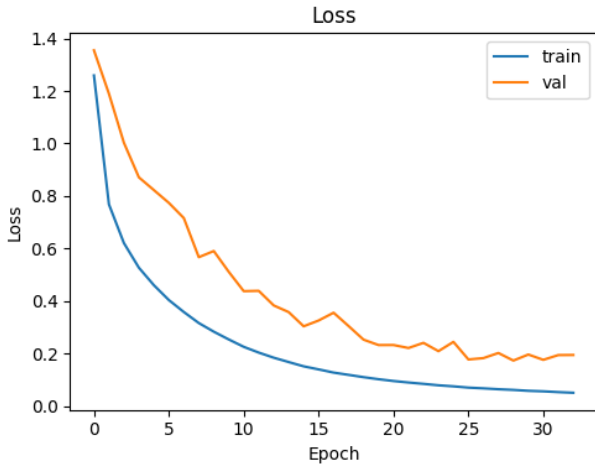


Figure 5. Training and validation loss

Classification Report			
Class	Precision	Recall	F1-score
0	1.00	1.00	1.00
1	1.00	0.94	0.97
2	0.87	1.00	0.93
3	1.00	0.69	0.82
4	0.96	0.98	0.97
5	1.00	1.00	1.00

Table 3. Performance metrics after testing

Table 4. Confusion matrix for hip-hop move classification

	Class	Pred					
		BS	CP	KBC	PdB	SM	WT
True	BS	8	0	0	0	0	0
	CP	0	29	0	0	2	0
	KBC	0	0	34	0	0	0
	PdB	0	0	4	9	0	0
	SM	0	0	1	0	44	0
	WT	0	0	0	0	0	16

underrepresented in this dataset. This suggests that while the features are simple and interpretable, they may not fully capture the nuances that dancers use to distinguish closely related movements.

Despite these limitations, the model performed well on the test set as well as in the real-time program. It could continuously predict the current move from a sliding window of frames. The small model size and the one-dimensional timeseries architecture make real-time use on a standard laptop feasible. This makes this system promising as a low-cost, markerless MoCap tool for educational and artistic applications. For example to give beginner dancers instant feedback on whether they are performing a move correctly, or to automatically annotate hip-hop dance segments in longer videos.

However, this project also has some clear constraints. The dataset is relatively small and mainly consists of recordings of one experienced dancer, with a small number of clips from one additional, non-expert dancer. The recordings were made in a controlled setting with mostly standard executions of each move at approximately four-count tempo. This limits the diversity of styles, body types and performance conditions. This makes the model’s ability to generalize to other dancers and more varied real-world scenarios uncertain. Future work should therefore collect data from more dancers with different body types and experience levels, include a wider range of tempos, include a larger movement vocabulary, and potentially extend the features with joint velocities and/or angles.

7. CONCLUSION

This project demonstrates that a small vocabulary of six hip-hop moves can be recognized from camera-based, markerless MoCap-data using simple, low-dimensional and human-interpretable 3D joint trajectories in combination with a shallow 1D-CNN model, with strong perfor-

and feet). This data was centralized by the mid-hip and normalized by shoulder-width to obtain translation- and scale-invariant trajectories. The features were designed and chosen based on visual inspection and dance experience, and by testing different normalization strategies. The system achieves a test accuracy of 0.95, with high precision and high recall for most classes. From the perspective of MoCap, this suggests that carefully designed skeletal features and basic timeseries modeling are sufficient to solve this particular movement classification task, without resorting to more complex 2D- or 3D-CNN architectures. However, there are some fundamental weaknesses in the system that are revealed by the classification report and confusion matrix. PdB is frequently misclassified as KBC, and SM is sometimes confused as KBC as well. From a MoCap perspective this is reasonable. These movements have similar lower body structure and step patterns. The current feature set emphasizes shoulders, elbows, wrists, hips, knees, ankles, heels and feet. As a result, certain differences in weight transfer and torso orientation may be

mance on the test set. This suggests that carefully designed skeletal features and basic timeseries modeling can solve this specific movement classification task without resorting to more complex 2D- or 3D-CNN architectures. At the same time, the current dataset is small and dominated by one experienced dancer, which limits the diversity of styles, body types and performance conditions and leads to confusions between, for instance, PdB and KBC. Future work should therefore expand the dataset and movement vocabulary and explore extended feature sets, while further developing educational and artistic applications such as automatic annotation of hip-hop segments in longer videos and real-time feedback for learners.

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